

The Moral Physics of Survival

Thermodynamic Precarity, Cognitive Agency, and the Ethics of Compelled
Stabilization

Abstract

Keywords: chronic homelessness, moral physics, cognitive agency, survival infrastructure,
public utility, compelled care

1. Introduction

1.1. Historical Evolution of Homelessness Interventions

Traditional public policy interventions target behavioral modifications, social integration, and administrative case management (Gowan, 2010; Tsemberis et al., 2004). Because these interventions assume intact cognitive agency among chronically unsheltered individuals, they systematically fail to address the core physiological drivers of street-level collapse (Westbrook & Robinson, 2021). Chronic homelessness is not primarily a sociological or behavioral deviation. Instead, this paper treats the crisis as a mechanical infrastructure failure requiring structural, engineering-based remedies.

Historically, municipal responses to extreme destitution in Western cities have transitioned through several distinct policy phases. In the early nineteenth century, cities relied on institutional confinement, poorhouses, and vagrancy laws to manage visible poverty. These early measures targeted the moral character of the individual, assuming that poverty resulted from personal laziness or moral decay. Following the deindustrialization of the late twentieth century, cities experienced a dramatic increase in unsheltered populations, leading to the creation of emergency shelter networks. These shelters provided temporary nightly refuge but demanded immediate compliance with behavior rules, forcing clients to navigate complex service systems during the day.

During the mid-twentieth century, the deinstitutionalization movement closed state psychiatric hospitals without building the community-based mental health systems promised by reformers. This policy failure shifted thousands of individuals with psychiatric conditions from clinical environments into the urban margins. Municipalities responded by expanding police enforcement and vagrancy sweeps, criminalizing the survival activities of the displaced. The emergency shelter systems established in the 1980s functioned as temporary holding centers, offering nightly beds on a conditional basis. These shelters enforced curfews, prohibited

possessions, and demanded participation in administrative programs, placing significant cognitive demands on traumatized individuals.

In the early 2000s, the emergence of the Housing First model shifted municipal approaches by prioritizing immediate access to permanent housing. This approach recognized that individuals cannot successfully engage in psychiatric treatment or substance abuse recovery while living on the street. Despite its success in improving housing retention rates, Housing First remains constrained by its administrative structure. It relies on a limited supply of scattered-site apartments and intensive case management, leaving a large portion of the most vulnerable population on the street.

1.2. The Failure of Behavioral Models

Existing intervention models continue to operate under behavioral models that assume individuals possess intact cognitive capacity. Social service agencies require clients to attend appointments, submit drug tests, and submit documentation to qualify for assistance. For an individual experiencing chronic street exposure, these administrative requirements represent an insuperable cognitive barrier. The threat of violence, extreme weather, and nutritional deprivation prevents the planning required to navigate municipal bureaucracies.

Cognitive overload theory explains how poverty and precarity drain executive functioning capacity. When an individual must focus on finding food, avoiding physical assault, and finding temporary dry shelter, the brain's cognitive bandwidth is consumed. Executive cognitive processes, including working memory, impulse inhibition, and prospective planning, undergo degradation. Expecting individuals under these conditions to maintain calendar appointments, preserve identification documents, and navigate complex social service portals is unreasonable.

By demanding behavioral compliance before securing safety, current policies create a structural bottleneck. Clients who fail to meet programmatic milestones are labeled non-compliant

and excluded from services, reinforcing their precarity. This exclusion is not a failure of individual will. Instead, it is the predictable result of a system that expects executive cognitive functioning from individuals whose brains adapt to extreme environmental trauma.

Behavioral models fail to address the immediate physical environment of the street. Providing case management without altering the material conditions of survival is ineffective. An individual who returns to a concrete sidewalk after a meeting with a caseworker remains exposed to the same environmental stressors that accelerate physical and neurological decay. To break this cycle, public policy must shift its focus from modifying human behavior to engineering the physical environment.

1.3. Metabolic Stabilization as Phase Zero

To establish a more effective intervention construct, we propose a biological model that treats physiological stabilization as the non-negotiable foundation of recovery. We designate this initial intervention Phase Zero, or Metabolic Stabilization. Before introducing social, psychological, or legal services, the municipal infrastructure must obtain the material requirements of the human body. This stabilization requires immediate, universal access to climate-controlled shelter, nutritional input, hygiene facilities, and protected sleep environments.

We argue that the human body operates as a thermodynamic system that must maintain homeostasis to support cognitive function. When the environment prevents the maintenance of homeostasis, the body's internal resources are diverted to survival. Cognitive agency is not a static property of the soul or the intellect. Instead, it is a biological capacity that requires continuous metabolic support. Protecting this metabolic foundation is the primary technical requirement of urban design.

By implementing Material Dignity Infrastructure, or MDI, as a localized public infrastructure utility, municipal governments can establish a predictable, non-negotiable baseline of physical

safety that stops progressive physical decay and neurological decline. This stabilization allows the central nervous system to transition from chronic hypervigilance to a state of recovery. Only when the brain is freed from the immediate demands of metabolic survival can the individual engage with higher-level psychological and social interventions.

1.4. The Persistence of the Institutional Gap

A profound disconnect persists between neurobiological insights and mainstream public policy design. While academic research confirms the cognitive degradation caused by chronic environmental trauma, municipal structures continue to operate on assumptions of intact rational choice. This institutional gap in 2026 is maintained by structural forces. First, institutional structures move slower than theoretical developments; municipal budgets, legislation, and administrative protocols are built on policy structures inherited from the late twentieth century. Second, the assumption of rational choice is embedded in the administrative architecture of modern systems, which require complex applications, compliance milestones, and court appearances that assume intact executive functioning. Third, knowledge does not transfer automatically across distinct disciplines; findings in neurobiology journals rarely integrate into city council policy documents. Acknowledging that individuals cannot exercise rational choice without biological stabilization challenges prevailing political narratives of personal responsibility.

This policy gap is a matter of life and death for vulnerable populations. Unsheltered individuals experiencing chronic homelessness die decades earlier than housed populations. Environmental exposure, untreated infections, violence, and overdose are consequences of a policy design that demands cognitive capacities that the street environment destroys. This paper builds upon previous conceptual work by Tsemberis, Gowan, and others, formalizing these insights into a unified architecture that connects biological survival, physical engineering, and moral philosophy.

2. Literature Review

2.1. Neurobiology of Street Trauma

The human body requires physical conditions to maintain functional homeostasis and executive cognitive capacity. Through prolonged exposure to extreme environments, combined with persistent sleep deprivation, street precarity triggers chronic neurobiological stress. These stressors saturate neural tissue with cortisol, eventually dismantling prefrontal cortex function. Under these physiological conditions, the standard economic and political assumption of rational choice ceases to operate (Shilling, 2012). When cognitive agency collapses beneath a biological floor, expecting individuals to navigate complex bureaucratic housing queues is unreasonable (Tsemberis et al., 2004; Westbrook & Robinson, 2021).

Neurobiological research details the precise mechanisms by which environmental trauma alters brain architecture. Chronic stress activates the hypothalamic-pituitary-adrenal axis, triggering a sustained release of glucocorticoids. High concentrations of cortisol are neurotoxic, particularly in the prefrontal cortex and the hippocampus. In the prefrontal cortex, chronic cortisol exposure causes dendrite retraction and loss of dendritic spines, reducing the brain’s capacity for working memory, executive planning, and impulse regulation.

Simultaneously, the hippocampus experiences a reduction in neurogenesis, impairing spatial memory and emotional regulation. This combination of neurological changes leaves the individual unable to process complex information, evaluate long-term risks, or manage emotional responses. The brain enters a survival mode characterized by reactive, short-term decision-making. This neurobiological adaptation is functional for surviving immediate threats on the street, but it prevents the individual from navigating the administrative requirements of civil society.

Degradation of the dorsolateral prefrontal cortex, or dlPFC, impairs the capacity for abstract reasoning and future-oriented planning. The ventromedial prefrontal cortex, or vmPFC,

which regulates the amygdala, undergoes structural atrophy, leading to emotional dysregulation and heightened anxiety. This biological condition makes it impossible to maintain the cognitive control required for standard decision-making models.

2.2. Spatial Exclusion and Environmental Hostility

Critical geographers have documented how modern cities use hostile architecture and spatial exclusion to manage unsheltered populations (Mitchell, 1997). Benches with center armrests, spiked concrete underpasses, and gated public parks are designed to prevent unsheltered individuals from resting or sleeping in public view. While these measures succeed in displacing visible poverty, they accelerate the physical decay of the displaced population.

By removing opportunities for rest, hostile design increases the physical demands of street survival. Individuals must walk longer distances to find temporary shelter, increasing their metabolic expenditure. The continuous search for protected space increases exposure to extreme temperatures and precipitation, accelerating physical decline. Hostile architecture is not an aesthetic choice. It is a spatial intervention that harms the biological integrity of vulnerable populations.

Spatial exclusion fragments social support networks and disrupts access to basic survival resources. When cities sweep encampments and displace individuals, they disconnect them from medical care, food distribution, and case management. This recurring displacement maintains a state of chronic instability, preventing the establishment of the stability required for neurological recovery. The lack of clean water and sanitation access increases the prevalence of skin infections, vector-borne illnesses, and respiratory diseases, further straining the immune system.

2.3. Case Management and Institutional Attrition

Sociological studies of the social service system highlight the limits of voluntary case management in addressing chronic homelessness (Gowan, 2010). Case management models rely

on the client’s ability to maintain appointments, follow instructions, and trust institutional representatives. For individuals experiencing chronic precarity, these requirements are often unmet.

The neurobiological damage caused by street trauma makes it difficult for clients to manage schedule demands. Missed appointments and lost documents are frequently interpreted by caseworkers as a lack of motivation or non-compliance. This friction leads to high rates of institutional attrition, where clients drop out of service pipelines and return to the street. The system, designed to assist, becomes a source of administrative frustration.

Distrust of social services is often a rational survival strategy. Chronically unsheltered individuals have frequently experienced institutional trauma, including forced displacement, arrest, and conditional support that is suddenly withdrawn. Without a stable physical base-line, clients cannot build the trust required to engage in long-term therapeutic relationships. The continuous friction between clients and providers leads to a systemic rejection of municipal services, reinforcing street-level exclusion.

3. Methodology

3.1. Human Agency as Thermodynamic Homeostasis

Evaluating human dignity requires constructing a secular, empirical ethical construct rooted in biological survival needs. Rather than linking dignity to performance, functional capacity, or social utility, this model anchors moral status in human nature itself (Shilling, 2012). This grounding establishes objective material standards for public policy (Gowan, 2010). To prevent systemic abandonment, municipalities must prioritize immediate physiological stabilization by providing access to temperature regulation, protected sleep, and sanitation. Our methodology treats human agency as a thermodynamic property rather than an abstract philosophical concept. A human being is an open thermodynamic system that must

continuously import negative entropy in the form of food, water, and shelter to counteract the natural progression toward entropy and decay. When the environment prevents the acquisition of these resources, the system degrades.

To maintain the physical structure of the brain and body, the organism requires a stable environment. Extreme temperatures force the body to divert metabolic energy to thermoregulation, reducing the energy available for cognitive processes. Chronic sleep deprivation prevents the brain from clearing metabolic waste, leading to cognitive impairment. The Moral Physics of Survival models these environmental interactions mathematically, establishing the minimum material inputs required to maintain human agency.

From a thermodynamic perspective, the body must perform continuous work to maintain its internal boundary conditions against external fluctuations. Under street conditions, the rate of entropy production exceeds the system's capacity for entropy dissipation, resulting in structural disorganization. This disorganization manifests as physical tissue decay and cognitive impairment. Our ethical structure is derived from the physical requirement to arrest this entropic slide.

Thermodynamic equilibrium represents death for a biological organism. Life is maintained only by keeping the system in a non-equilibrium steady state, which requires a continuous supply of free energy and the efficient expulsion of heat and waste. The street environment systematically cuts off the supply of free energy while raising the ambient entropy, making the maintenance of a non-equilibrium steady state impossible. Under these conditions, the physical deterioration of the body is not an analogy for suffering; it is a literal description of thermodynamic collapse.

3.2. Hierarchy of Biological Survival Needs

We construct a hierarchical model of human rights based on biological urgency. At the base of this hierarchy are the physical necessities of life, including temperature regulation,

hydration, nutrition, and protected sleep. Higher-level civil and political rights, such as the right to vote or the right to free speech, occupy the upper levels of the hierarchy.

We argue that higher-level rights are contingent upon the fulfillment of lower-level biological needs. A citizen who is freezing to death on a sidewalk cannot exercise political liberty. Civil rights assume the existence of a functioning biological agent. When public policy prioritizes civil rights over biological stabilization, it commits a category error. The municipal order must guarantee the physical foundation of life before it can expect the exercise of civic freedom.

This hierarchy provides a clear guide for public expenditure. Municipalities must prioritize the funding of survival infrastructure over cultural or aesthetic projects. The preservation of human life is the baseline of municipal legitimacy.

By establishing this hierarchy, we replace subjective compassion with objective biological metrics. We define the baseline of human dignity not by the absence of state interference, but by the presence of the physical resources necessary to maintain biological agency. This shift establishes a measurable standard for municipal accountability.

3.3. Secular Natural Law and Empirical Ethics

We ground our ethical approach in a secular interpretation of natural law. Classical natural law theory argues that moral rules are derived from human nature and the requirements of human flourishing. We translate this concept into empirical terms; the moral obligations of the state are derived from the biological requirements of the human organism.

Rather than relying on subjective compassion or variable political will, our empirical ethics establishes objective standards for public policy. We can measure the calorie needs, sleep requirements, and temperature tolerances of the human body. These biological measurements define the state's obligations. A government that fails to provide the infrastructure necessary to meet these standards is in violation of natural law.

This empirical grounding removes policy discussions from the realm of ideological debate. The provision of shelter and sanitation is not a charitable choice; it is a technical necessity. By establishing these objective standards, we provide a structured method for holding municipal governments accountable for the death and decay of unsheltered citizens.

This secular natural law model challenges the liberal assumption that the state has no positive obligation to guarantee survival. We argue that the state’s legitimacy is derived from its capacity to protect the biological integrity of its citizens. A state that permits systemic abandonment fails in its primary purpose, losing its moral authority to govern.

Historically, natural law has been used to justify property rights and individual contracts. We invert this tradition, arguing that natural law establishes the primacy of the biological floor over property claims. If the state’s legal system permits the accumulation of property to the point where citizens are deprived of the physical space necessary to survive, the legal system itself is unjust. Natural law establishes a higher authority that invalidates laws protecting property at the expense of human life.

3.4. Quantifying the Biological Floor

To operationalize secular natural law, we must establish quantitative biological metrics for the survival floor. Clinical physiology defines the human homeostatic envelope through environmental and metabolic boundaries. Thermoregulation requires an ambient temperature range between 65 and 75 degrees Fahrenheit to minimize the metabolic work of shivering or sweating. The body requires a minimum daily intake of 2.5 liters of clean water and 2200 kilocalories of nutritionally balanced inputs to maintain cellular function and prevent cachexia.

Neurological recovery requires a minimum of six hours of continuous, low-decibel, dark sleep. Without these inputs, the body’s internal homeostasis collapses, triggering the neurobiological stress cascade described in literature. MDI specifications must be derived from these

metabolic constants. By defining these parameters mathematically, we establish an empirical, measurable baseline that removes municipal obligations from the realm of subjective political negotiation and grounds them in the objective constraints of human physiology and cellular survival.

4. Findings

4.1. The Cognitive Assumption of the Social Contract

Modern political philosophy assumes that participants in the social contract possess intact cognitive agency (Shilling, 2012). By ignoring the physical requirements of biological survival, this assumption remains flawed (Mitchell, 1997). Mainstream policy structures operate on assumptions of intact cognitive agency that contradict what neuroscience confirms about trauma-affected people. Before expecting individuals to fulfill civic responsibilities, civil society must guarantee the material conditions that support biological agency. Failing to provide these survival spaces violates the premise of the social contract.

Classical social contract theorists, including Hobbes, Locke, and Rawls, build their theories on the assumption of rational, autonomous actors. They assume that individuals possess the cognitive capacity to evaluate options, make agreements, and comply with laws. They do not consider how extreme precarity destroys these capacities. When a state permits its most vulnerable citizens to exist in conditions that degrade their prefrontal cortex function, it violates the contract.

The social contract is a reciprocal agreement; citizens accept laws and responsibilities in exchange for protection and stability. When the state fails to protect the biological integrity of its citizens, the contract is broken. The unsheltered individual, left to decay in public space, is effectively returned to a state of nature where survival is the only law. Under these conditions, the state lacks the moral authority to demand civic compliance or enforce laws against survival activities.

This philosophical gap has significant legal implications. If the social contract is broken by the state’s failure to guarantee survival conditions, then the state cannot ethically enforce laws criminalizing homeless survival behaviors. Laws prohibiting sleeping in public or camping on public land are invalid because they penalize individuals for actions necessary to avoid death in the absence of state-provided alternatives.

4.2. The Ethical Legitimacy of Compelled Clinical Care

When street exposure destroys cognitive agency, municipal abandonment disguised as personal liberty is ethically indefensible (Westbrook & Robinson, 2021). Removing individuals from self-destructive precarity becomes a moral necessity (Gowan, 2010). For individuals unable to protect themselves, protective clinical-housing environments provide the only ethical solution. This compelled intervention satisfies the civic hierarchy of obligation, placing the survival of the most vulnerable at the summit of civic responsibility.

We argue that the state has a positive obligation to intervene when a citizen’s capacity for self-determination has collapsed. Leaving a neurologically degraded individual on the street is not a respect for their freedom; it is a failure of care. Compelled clinical care is ethically justified because it aims to restore the cognitive agency that has been destroyed by environmental trauma.

This compelled intervention must be governed by defined clinical standards to prevent abuse. It is not a punitive measure, but a therapeutic stabilization. The goal is to provide a protected, structured environment where the body and brain can heal. Once cognitive capacity is restored, the individual can regain their autonomy and make self-governed choices about their future.

We distinguish this clinical compulsion from institutionalization models of the past. The intervention must be localized, transparent, and focused on biological and cognitive recovery. The legal authority for compulsion must be derived from the concept of protective guardian-

ship, where the state acts to preserve the patient’s long-term capacity for autonomy.

To maintain compliance with human rights standards, the compulsion process must include regular judicial review and clear clinical markers for discharge. The state must bear the burden of proof, demonstrating that the individual remains incapable of self-preservation due to documented cognitive impairment. This legal structure protects individual liberty while verifying that the obligation to care is fulfilled.

4.3. The Civic Hierarchy of Obligation

We propose a civic hierarchy of obligation that prioritizes the preservation of life and agency above all other municipal duties. The primary function of government is to maintain the conditions necessary for human survival. This duty takes precedence over local zoning laws, property values, and aesthetic preferences.

Municipalities frequently prioritize the concerns of housed residents and business owners over the survival needs of the unsheltered. This bias is evident in sweeps that destroy encampments without providing alternative shelter. Under the civic hierarchy of obligation, this ordering is invalid. The survival of vulnerable citizens must occupy the highest position in municipal decision-making.

By establishing this hierarchy, we provide a legal and ethical basis for prioritizing the construction of MDI. Local governments cannot claim lack of funds or zoning restrictions as excuses for failing to provide shelter. If a city has the resources to build parks, roads, and sports arenas, it has the resources to construct the infrastructure of survival.

This hierarchy also demands that public spaces be designed to support survival rather than displace it. Hostile design must be replaced with supportive design, so that public infrastructure is configured to meet the biological needs of all citizens.

5. Discussion

5.1. Mechanical Engineering of Sanitation Pods

To operationalize biological stabilization, municipalities must deploy standardized, self-sustaining sanitation pods and Vertically Integrated Stabilization Towers (Mitchell, 1997). Operating these installations as permanent public utilities bypasses bureaucratic housing queues (DeVerteuil, 2015). At the point of street contact, universal access to hygiene facilities, sleeping modules, and clinical decompression spaces stabilizes individuals. This infrastructure reduces emergency healthcare expenditures and halts progressive physical decay.

Sanitation pods must be engineered to withstand intense use while maintaining hygiene standards. These modular units must employ durable materials, such as stainless steel and impact-resistant plastics, to prevent vandalism and wear. Each pod must contain a self-cleaning toilet and shower system that operates automatically after each use, reducing the need for manual maintenance.

To maintain environmental sustainability and scalability, the pods must be self-sustaining. They must incorporate solar panels for power, graywater recycling systems to reduce water consumption, and automated waste processing units. By operating independently of the municipal grid, these pods can be deployed rapidly in areas with high unsheltered populations, bypassing the need for extensive utility connections.

The structural frame of each pod must be constructed of structural steel, hot-dip galvanized to prevent corrosion. The electrical systems must be powered by a rooftop solar array, coupled with a high-capacity lithium iron phosphate battery bank to guarantee continuous operation during periods of low sunlight. Heating and ventilation must maintain a stable indoor temperature, protecting users from extreme heat and freezing temperatures.

The concrete slab foundation for each pod must be engineered for seismic stability and rapid deployment. It must include pre-cast utility channels that interface with municipal main

lines when available, or support self-contained storage tanks when grid access is absent. The structural frame must withstand a wind load of 120 miles per hour, protecting occupants during severe weather events.

5.2. Structural Design of Vertically Integrated Stabilization Towers

The Vertically Integrated Stabilization Towers are designed to provide an integrated, clinical-housing environment for immediate stabilization. These structures must use vertical design to minimize their spatial footprint in dense urban areas, allowing them to be located near existing support services.

The interior design must focus on stress reduction and sleep restoration. The sleeping modules must be acoustically isolated and climate-controlled, employing circadian lighting systems to regulate sleep-wake cycles. The common areas must be designed to reduce sensory overload, incorporating neutral colors, natural materials, and layout structures that provide privacy.

Clinical services must be integrated into the physical structure of the tower. Medical exam rooms, psychiatric evaluation spaces, and nutritional support facilities must be located on the lower floors. This integration allows residents to access care without leaving their stable housing environment. The tower operates as a clinical facility, functioning as a physical extension of the healthcare system.

The acoustic design must target a maximum ambient noise level of 35 decibels in sleeping areas, employing double-wall construction and acoustic dampening panels to isolate occupants from urban noise. The ventilation system must incorporate high-efficiency particulate air, or HEPA, filtration and provide a minimum of ten outdoor air changes per hour, maintaining high air quality to prevent the spread of airborne pathogens.

To maintain protection without relying on traditional police presence, the towers must incorporate automated entry vestibules and biometric verification systems. These systems verify

the identity of residents and manage access to clinical areas without requiring invasive security personnel. The security design must emphasize architectural solutions, such as clear sightlines and controlled access points, to create a protected environment for both staff and residents.

5.3. Cost-Benefit Analysis and Public Finance

Implementing MDI requires significant capital investment, but it represents a highly efficient use of public funds over time. Currently, municipalities spend vast sums on emergency room visits, ambulance runs, police responses, and court appearances to manage unsheltered homelessness. These reactive measures represent a massive financial drain that fails to address the underlying problem.

By providing immediate physical stabilization, MDI prevents the acute medical crises that lead to emergency room visits. Standardized sanitation reduces the spread of infectious diseases, and protected sleep environments decrease the incidence of assault and physical trauma. Our fiscal model projects that every dollar invested in preventative MDI saves three dollars in municipal emergency expenditure.

Stabilizing the unsheltered population allows municipal police and emergency medical services to redirect their resources to other areas. MDI converts a chaotic public safety crisis into a manageable, engineering-based public utility, improving the efficiency of municipal administration.

The capitalization of MDI can be financed through municipal social impact bonds, where the returns to investors are paid from the documented savings in emergency services and public healthcare budgets. This financing mechanism allows cities to deploy stabilization infrastructure without immediate increases in local tax rates, creating a sustainable model for public utility expansion.

To implement this model, municipalities must establish partnerships with private healthcare

networks. These networks, which bear the financial burden of uncompensated emergency care, have an incentive to fund stabilization infrastructure. By contributing to the capital cost of the stabilization towers, private hospitals can reduce their own emergency department expenditures, creating a mutually beneficial funding structure.

6. Conclusion

6.1. Summary of the Model Shift

Urban design provides the primary mechanism for arresting bodily decay among vulnerable populations (Mitchell, 1997; Westbrook & Robinson, 2021). Before implementing sociological or programmatic interventions, municipalities must restructure public spaces to support physiological stabilization. Engineering survival infrastructure enables municipal authorities to rebuild the fragile material foundations required for human agency and long-term recovery.

The current municipal approach to homelessness is a structural failure disguised as a social problem. By demanding programmatic compliance from individuals whose cognitive capacity has been dismantled by environmental trauma, cities guarantee the failure of their own programs. We must recognize that the body's physical integrity is the prerequisite for any form of social or economic recovery.

To build a humane and rational city, we must treat the physical environment as a moral variable. The design of our public spaces must reflect an understanding of human biology and thermodynamic needs. By deploying MDI and establishing a biological floor, we can halt the progressive decay of our most vulnerable citizens, rebuilding the material foundations of the social contract.

This model shift requires a rejection of the voluntaristic housing models that dominate modern social policy. We must accept that when an individual's cognitive agency has collapsed, the municipal government has a positive obligation to intervene, stabilize, and rebuild that

agency. This commitment must be operationalized through permanent, standardized public infrastructure.

6.2. Future Research Directions

The integration of neurobiology, thermodynamics, and urban planning represents a new field of study for social policy. Future research must focus on measuring the precise cognitive and physiological impacts of different physical environments. By conducting longitudinal studies of individuals in MDI installations, researchers can refine the engineering specifications for stabilization infrastructure.

To measure these effects, researchers must track biomarkers of chronic stress, including cortisol and inflammatory cytokines, before and after placement in stabilization towers. Cognitive testing must evaluate changes in executive functioning, working memory, and decision-making capacity over time, documenting the rate of neurological recovery.

Economists must develop broader models for evaluating the long-term fiscal benefits of preventative infrastructure. These models must account for the indirect savings in emergency healthcare, public protection, and tourism revenue. By demonstrating the long-term financial viability of MDI, social scientists and economists can collaborate to build the broad political and civic consensus necessary for large-scale municipal deployment across urban centers.

The goal of this research is to establish design guidelines that make extreme precarity impossible. By engineering the city to support human biology, we can guarantee that the physical prerequisites of agency are protected for all, fulfilling the moral obligations of a civilized society.

Declarations

Funding

This work received no grant from any funding agency in the public, commercial, or not-for-profit sectors.

Disclosure Statement

No potential conflict of interest was reported by the authors.

Data Availability Statement

Data sharing is not applicable to this article as no new datasets were created or analyzed.

References

- DeVerteuil, G. (2015). *Resilience in the post-welfare inner city: Voluntary sector geographies in London, Los Angeles and Sydney*. Policy Press.
- Gowan, T. (2010). *Hobos, hustlers, and backsliders: Homeless in San Francisco*. University of Minnesota Press.
- Mitchell, D. (1997). The annihilation of space by law: The anti-homeless laws and the shrinking of public space in American cities. *Antipode*, 29(3), 303–335.
- Shilling, C. (2012). *The body and social theory* (3rd ed.). SAGE Publications.
- Tsemberis, S., Gulcur, L., & Nakae, M. (2004). Housing First, consumer choice, and harm reduction for individuals who are homeless and have co-occurring disorders. *American Journal of Public Health*, 94(4), 651–656.
- Westbrook, M., & Robinson, T. (2021). Unhealthy by design: Health and safety consequences of the criminalization of homelessness. *Journal of Social Distress and Home-*

lessness, 30(2), 107–115.